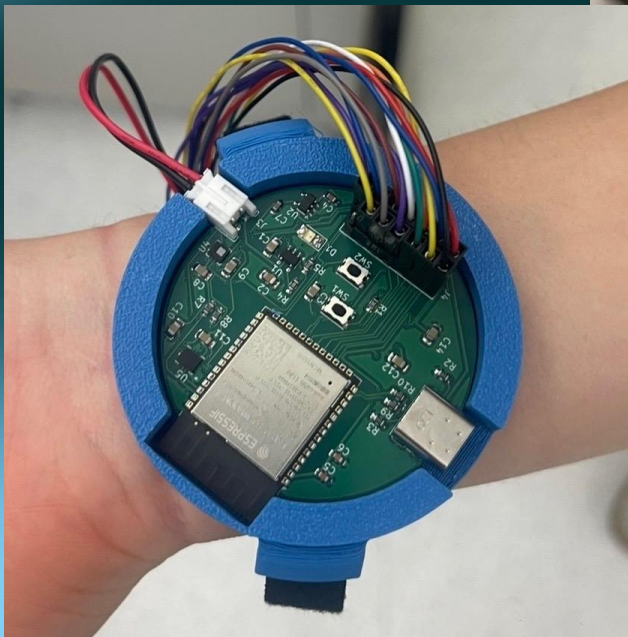


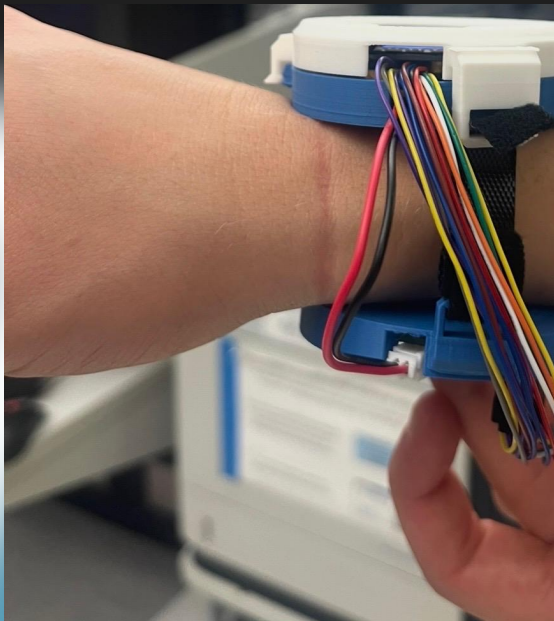
stryde

Biometric smartwatch for health and fitness goals

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Natalie Wang,
Siddharth
Veerapaneni, and
Zoe Lahanas



Overview



Our product aims to deliver on-demand statistics of your day to day activities, tracking various types of motion, biometric data , and environmental factors.

Specs

- 3.7V battery for all day wear
- Bluetooth compatible
- Touch screen display
- PCB Status LED to monitor power delivery
- Lightweight space efficient design



BOM

Qty (for 1 board)	Manufacturer Part #	Digi-key Part#/SKU	Mouser Part#/SKU	Price/unit (\$)	Purchase URL	Reference Designator	Description (include component value if applicable)	Datasheet Link
1	ESP32-S3-WROOM-1-N16R8	5407-ESP32-S3-WROOM-1-N16R8CT-ND		0	https://www.digik	U3	ESP32	https://docum
2	C1206C106M8PACTU	399-C1206C106M8PACTUCT-ND		0.3	https://www.digikey.com/en/pro		10uF Capacitor	https://yageog
2	RC0603FR-0710KL	311-10.0KHRCT-ND		0.1	https://www.digik	R1	10 kOhm resistor	https://yageog
1	MCP9808T-E/MS	MCP9808T-E/MSCT-ND		1.4	https://www.digik	U6	skin temp sensor	https://ww1.m
4	RMCF0603FT4K70	RMCF0603FT4K70CT-ND		0.1	http://digikey.com	R7, R8	I2C pull-ups (4.7k resistor)	https://www.se
1	LPS25HBTR	497-15070-2-ND		4.5	https://www.digik	U4	barometer	https://www.st
9	KGM21KR51H104KU	478-KGM21KR51H104KUCT-ND		0.1	https://www.digik	C3, C8, C9, C10	100 nF capacitor	https://datashe
4	C0603C105K4RACTU	399-C0603C105K4RACTUCT-ND		0.08	https://www.digik	C4, C5, C7	AP2112K input + output (1u ca	Datasheet
1	MSASQ21GSB5103KTNA01	587-MSASQ21GSB5103KTNA01CT-ND		0.18	https://www.digik		10 nF capacitor	https://mm.dig
1				26.99	https://www.ama		LED Touchscreen Display	
1	USB4085-GF-A	2073-USB4085-GF-ACT-ND		0.89	https://www.digik	J2	usbc charging	https://gct.co/f
1	MCP73831T-2ACI/OT	MCP73831T-2ACI/OTCT-ND		0.76	https://www.digik	U1	battery charging chip	https://ww1.m
1	AP2112K-3.3TRG1	AP2112K-3.3TRG1DICT-ND		0.22	https://www.digik	U2	3.3V regulator	https://www.d
1	S2B-PH-K-S	455-1719-ND		0.11	https://www.digik	J3	Battery connector	Datasheet
1	1578	1528-1841-ND		7.95	https://www.digik		LIPO Battery	
1	LSM6DSO32TR	497-LSM6DSO32TR-ND		3.5	https://www.digik	U5	IMU	https://www.st
4	RMCF0603FT5K10	RMCF0603FT5K10CT-ND		0.1	https://www.digik	R3, R2	USB-C CC1/CC2 pull-downs (5	https://www.se
2	RMCF0603FT2K00	RMCF0603FT2K00CT-ND		0.1	https://www.digik	R4	MCP73831 RPROG (2k resistor	https://www.se
2	RMCF0603FT1K00	RMCF0603FT1K00CT-ND		0.1	https://www.digik	R5	Charge LED current limiter (1k	https://www.se
3	RMCF0603FT22R0	RMCF0603FT22R0CT-ND		0.1	https://www.digik	R9, R10	USB D+/D- series resistors (22	https://www.se
3	CL21A475KPFNNNF	1276-CL21A475KPFNNNFC		0.1	https://www.digik	C1, C2	4.7uF capacitor	https://weblib
2	LTST-C230KRKT	160-1457-2-ND		0.15	https://www.digik	D1	Red LED	https://mm.dig
2	KMR221G LFS	108-KMR221GLFSCT-ND		0.61	https://www.digik	SW1, SW2	Tactile Switch SMD	https://www.lit
1	67996-114HLF	609-3491-ND		1.08	https://www.digik	J4	pin header 2x7	
2	C0603C104K3RACTU	399-C0603C104K3RACTUC		0.08	https://www.digik	C6	.1 uF capacitor	https://yageog

Schematic

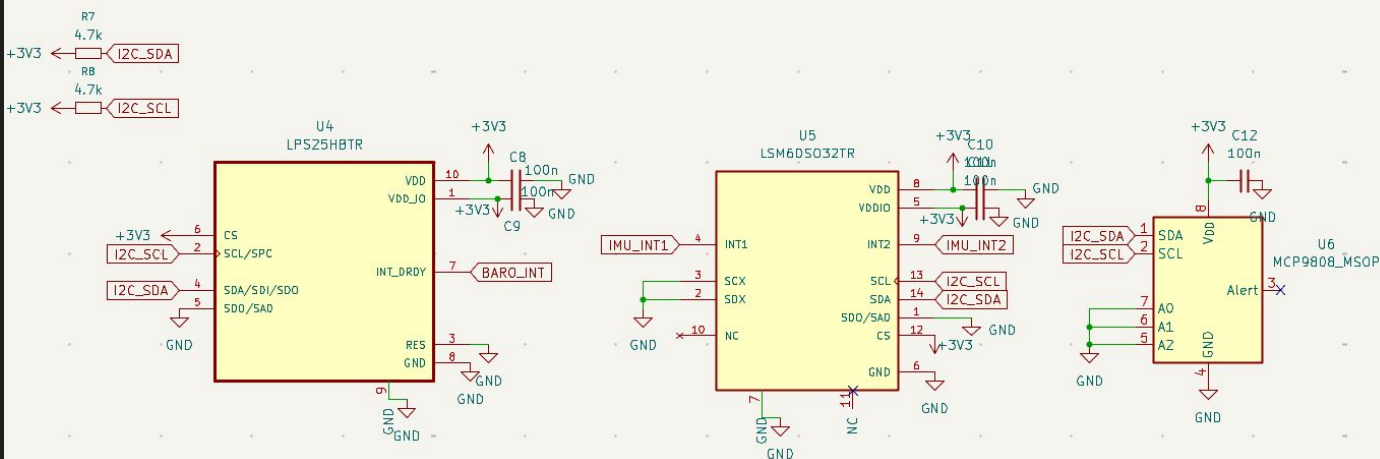
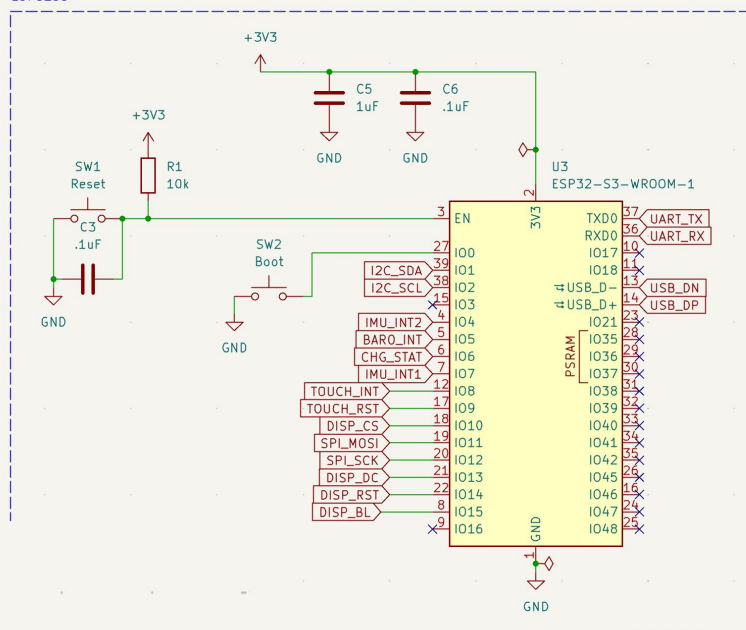
MCU

– BOOT and
RESET circuits

Sensors

- 2 pullup resistors: SDA and SCL to 3.3V to make the lines go high when none of the devices are pulling them low

ESP32S3



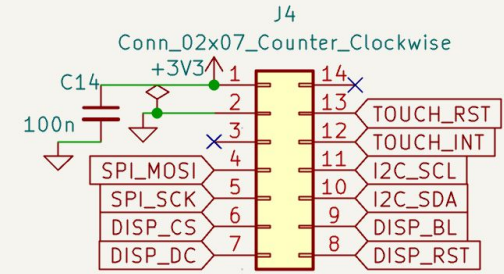
Schematic

Power

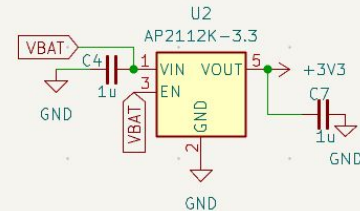
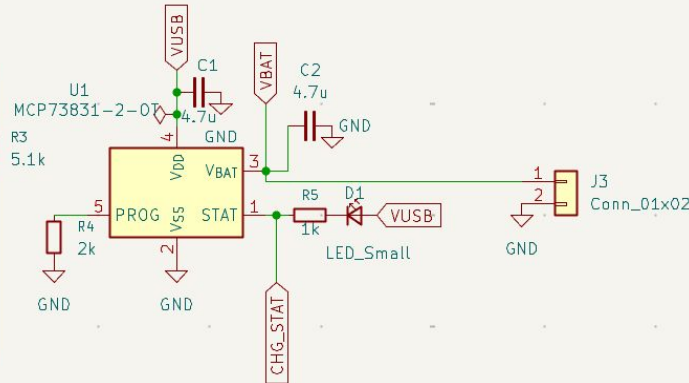
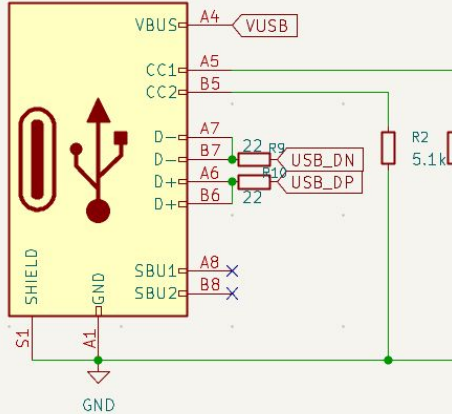
- CC pull-down resistors tell the USB-C host that the device wants power
- Series termination resistors on the data lines

Display

- Input and output capacitors
- R4 to set charge current
- Charge status indicator
- LDO Regulator drops V_{bat} to clean 3.3 V output

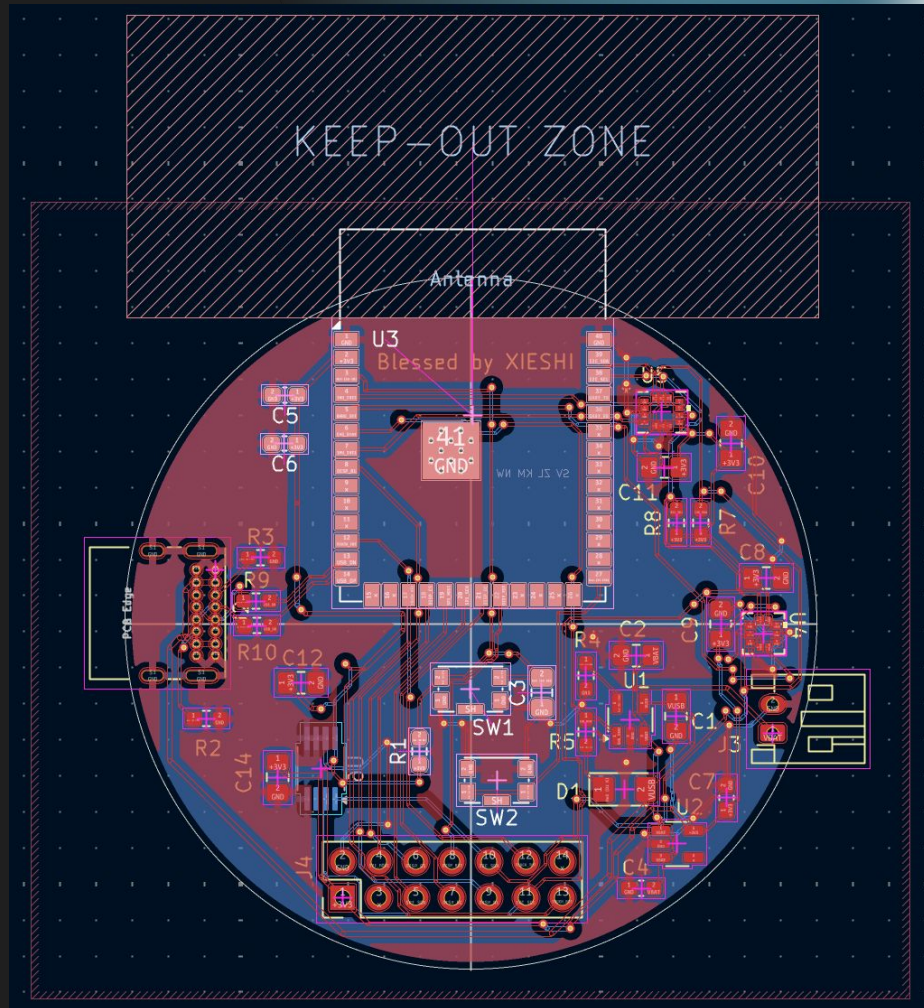


J2
USB_C_Receptacle_USB2.0_16P

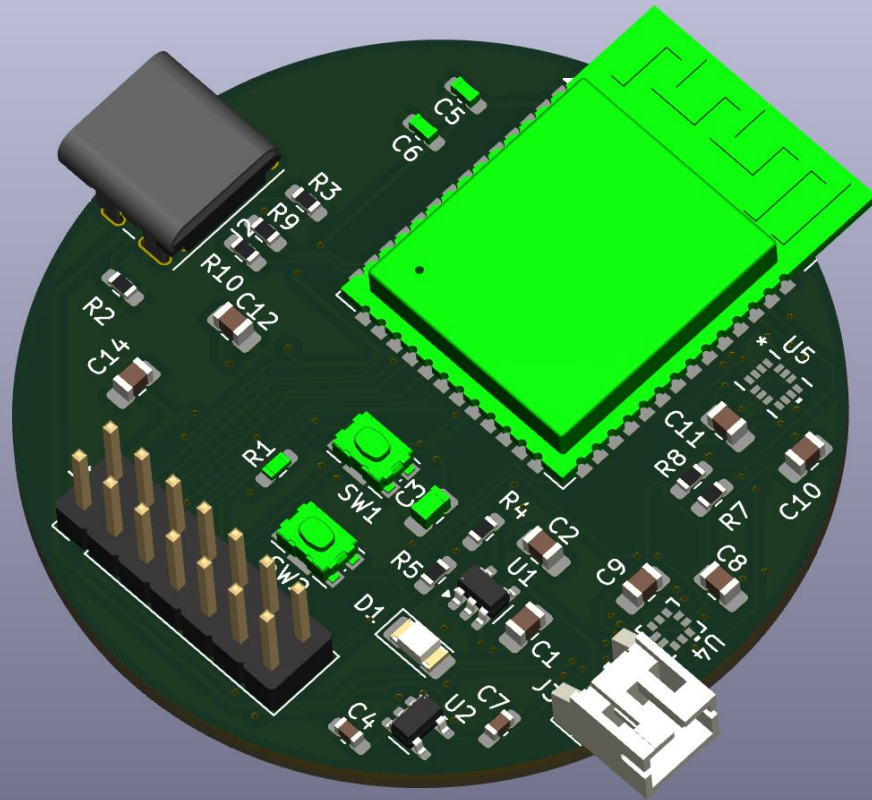


Layout

- Added heat relief in the form of vias
- Chose 2x07 connector instead of 1x14 for space
- Grouped majority of wires in specific parts of the board
- Used thicker wiring for power, battery, etc.



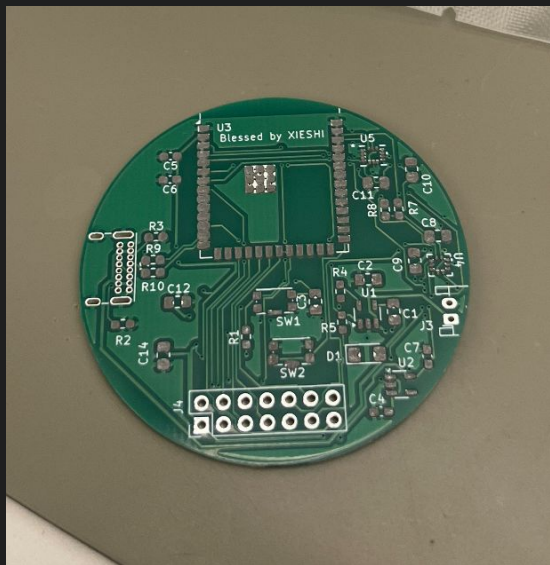
3D View



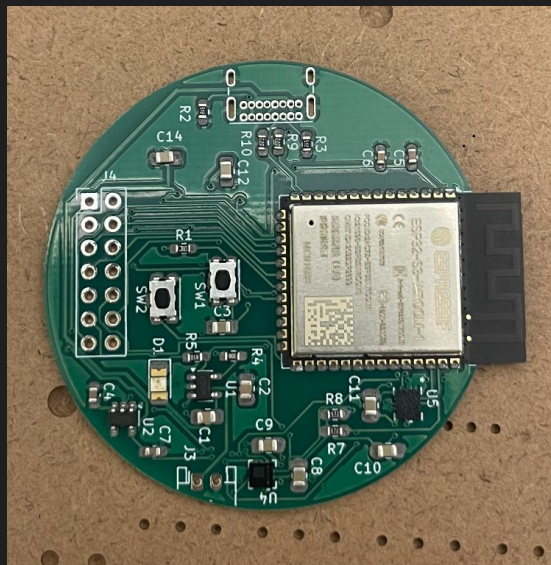
Reflow Soldering



Reflow Soldering



Solder paste applied

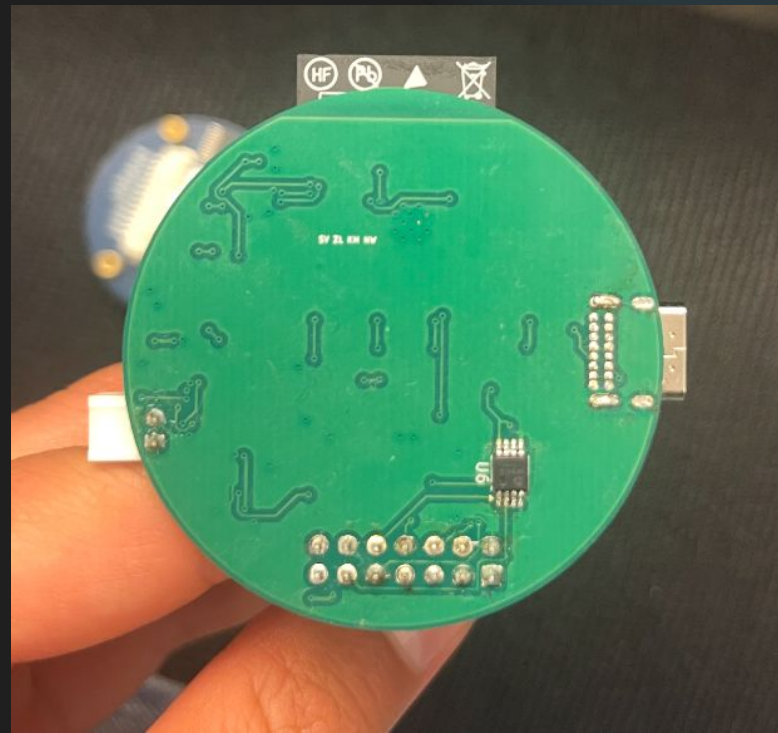
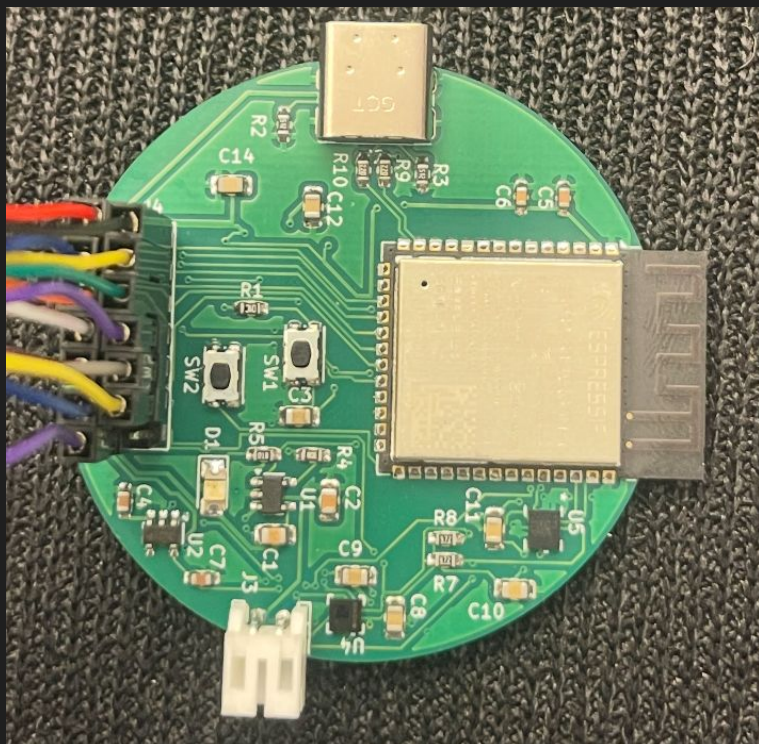


Components placed



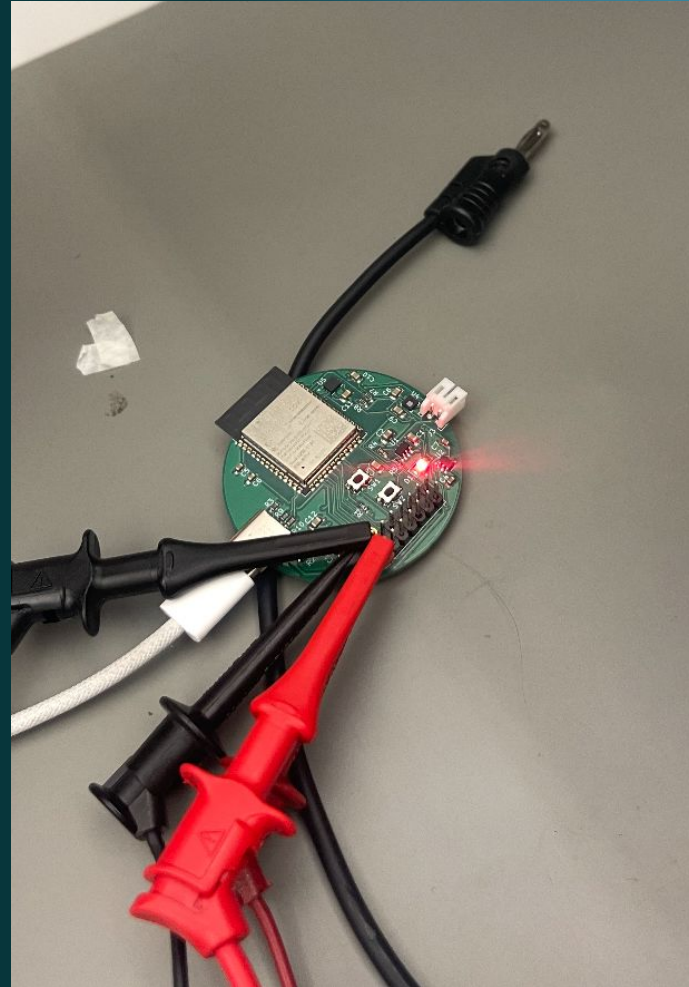
After heating in oven

Final Board



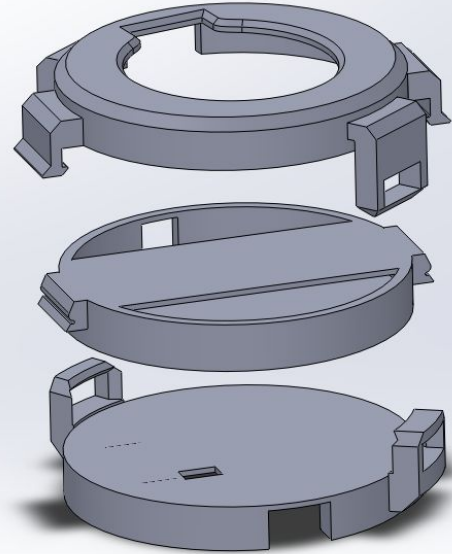
Testing

- Check that power works
- Check temperature
- Check if I2C connects to all ICs
- Check if screen turns on
- Check for shorts



Case Design

- Board accessibility (switches)
- Battery housing
- Screen mount
- Skin temp sensor contact



Firmware

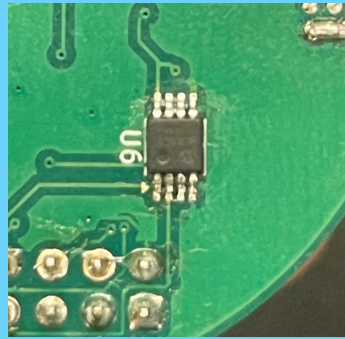
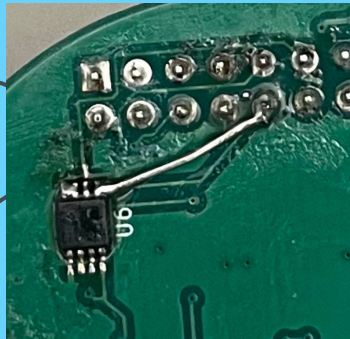
- Bluetooth
- Main screen: time, battery charge, bluetooth
- Second Screen: health data – skin temperature, step count
- Third Screen: environment – weather, altitude

Challenges

- Making the pcb a wearable size
- Soldering skin temperature sensor
- IMU and barometer not hand solderable
- Designing a case around battery/screen/board

Solutions

- ➡ Compact layout
- ➡ Second board
- ➡ Stencil/Reflow soldering
- ➡ Top/bottom of wrist case design



Future Improvements

- Add wifi
- More testing
- More compact – smaller battery, board, etc.
- Controls solely on the touch screen and not the board itself
- On-Device ML for exercise detection– dataset creation, augmentation, training, validation
- App development

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Thank you!
